LASER-CUTTER X900 Maintenance Manual

Chapter 1 – Machine Overview

The LASER-CUTTER X900 is an industrial fiber laser cutting machine used for precision cutting of stainless steel, aluminum, and mild steel sheets. The machine uses a high-power laser source, beam delivery optics, cooling system, and automatic positioning controller.

Error Code: LASER_HEAD_ALIGNMENT_FAILURE

Fault Description

The laser head alignment system has detected excessive deviation between the laser beam center and the focusing lens assembly.

This fault prevents accurate cutting and may reduce product quality.

Symptoms

- Cutting edges become uneven.
- Laser beam shifts away from programmed path.
- Automatic calibration fails.
- Position verification reports excessive offset.
- Cutting accuracy decreases.

Possible Causes

- Loose laser head mounting bolts.
- Contaminated focusing lens.
- Bent alignment bracket.
- Laser head impact during operation.
- Optical sensor misalignment.

Inspection Procedure

1. Power OFF the machine.

2. Inspect laser head mounting bolts.
3. Verify focusing lens cleanliness.
4. Check optical sensor position.
5. Inspect alignment bracket for bending.
6. Run laser calibration routine.

Repair Procedure

1. Tighten all laser head mounting bolts.
2. Replace damaged alignment bracket if bent.
3. Clean focusing lens using approved optical cleaning solution.
4. Recalibrate laser head alignment.
5. Verify beam position using calibration target.
6. Perform test cutting cycle.

Safety Precautions

- Disconnect machine power before servicing.
- Wear laser safety goggles.
- Never touch optical surfaces with bare hands.
- Keep cooling system operating during calibration.

Required Tools

- Torque wrench
- Alignment gauge
- Optical cleaning kit
- Precision hex key set
- Laser calibration target

Spare Parts

- Laser Head Assembly
- Focusing Lens
- Alignment Bracket
- Optical Position Sensor

Maintenance Notes

After completing repairs, perform three successful calibration cycles before returning the machine to production.

Record the calibration values in the maintenance log.

Normal alignment deviation must remain below 0.05 mm.